

AI-Assisted Kitchen Operations

A conversational ordering platform connecting customer messaging to structured domain actions, persistent kitchen workflows, live operational visibility, and reliable thermal printing.

Backend & AI Integration Developer

5

client establishments supported concurrently

~40%

reduction in manual operational work

4

API services integrated into the operating flow

MY CONTRIBUTION

- Owned backend and AI-integration features from requirements and flow design through testing and deployment.
- Built and evolved APIs and Model Context Protocol (MCP) tools for menus, orders, client context, and printing.
- Improved PostgreSQL and Prisma-backed domain models and migration workflows.
- Implemented voice interaction support and an Android/Kotlin bridge for Bluetooth ESC/POS printers.
- Strengthened automated unit, integration, and end-to-end testing around critical operational paths.

TECHNOLOGY

TypeScriptNode.jsREST APIMCPPostgreSQLPrisma ORMWhatsAppKotlin / AndroidESC/POSDockerCI/CDTDD

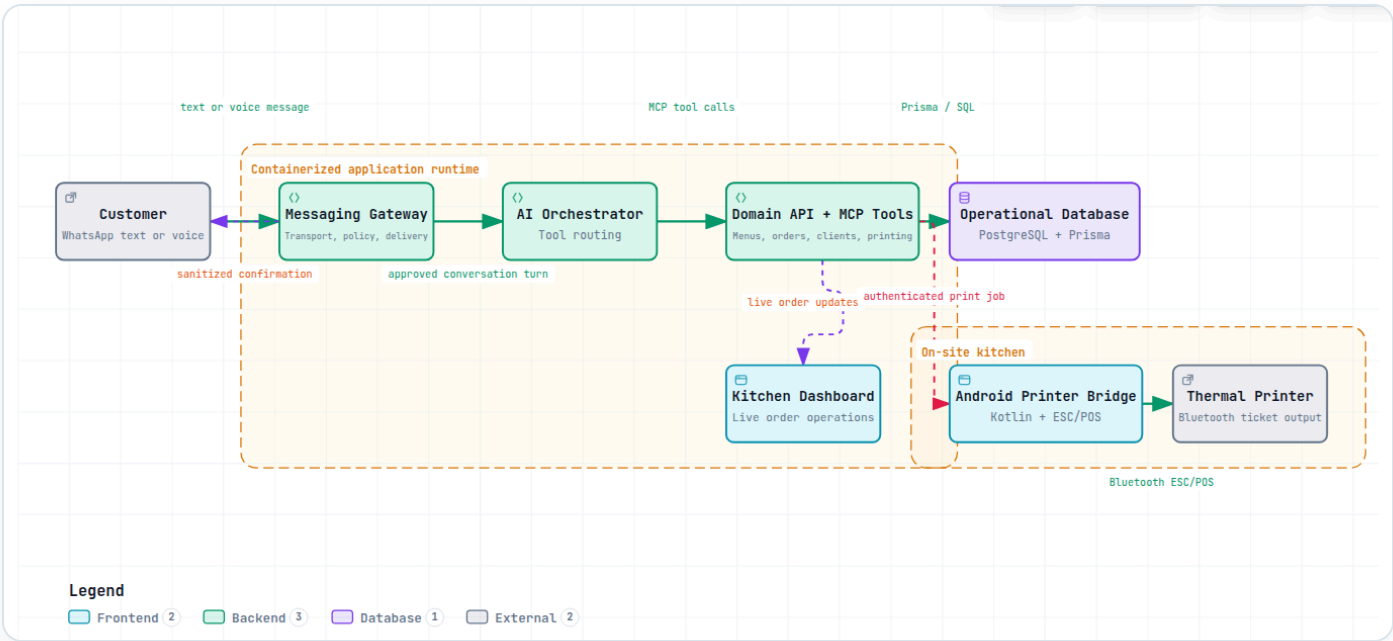
Confidentiality by design

This is a conceptual reconstruction based on my contribution. It contains no proprietary source code, customer data, credentials, private prompts, exact schemas, or internal identifiers.



Architecture and boundaries

The design separates conversation transport, AI tool orchestration, business state, live kitchen operations, and device-specific printing. This keeps each integration replaceable and reduces the amount of business logic embedded in the conversational layer.



Conceptual architecture; names and boundaries are intentionally generic.

Structured domain access

The AI orchestrator performs business actions through MCP tools instead of manipulating database state directly.

Persisted source of truth

Menus, orders, client context, and print state are validated by the domain API and persisted in PostgreSQL.

Hardware isolation

The Android bridge contains Bluetooth and ESC/POS concerns, keeping device behavior outside the backend runtime.

CONTRIBUTION MAP

APIs + MCP

Domain tools and integrations

Data layer

Models and migrations

Voice flow

Speech input and output

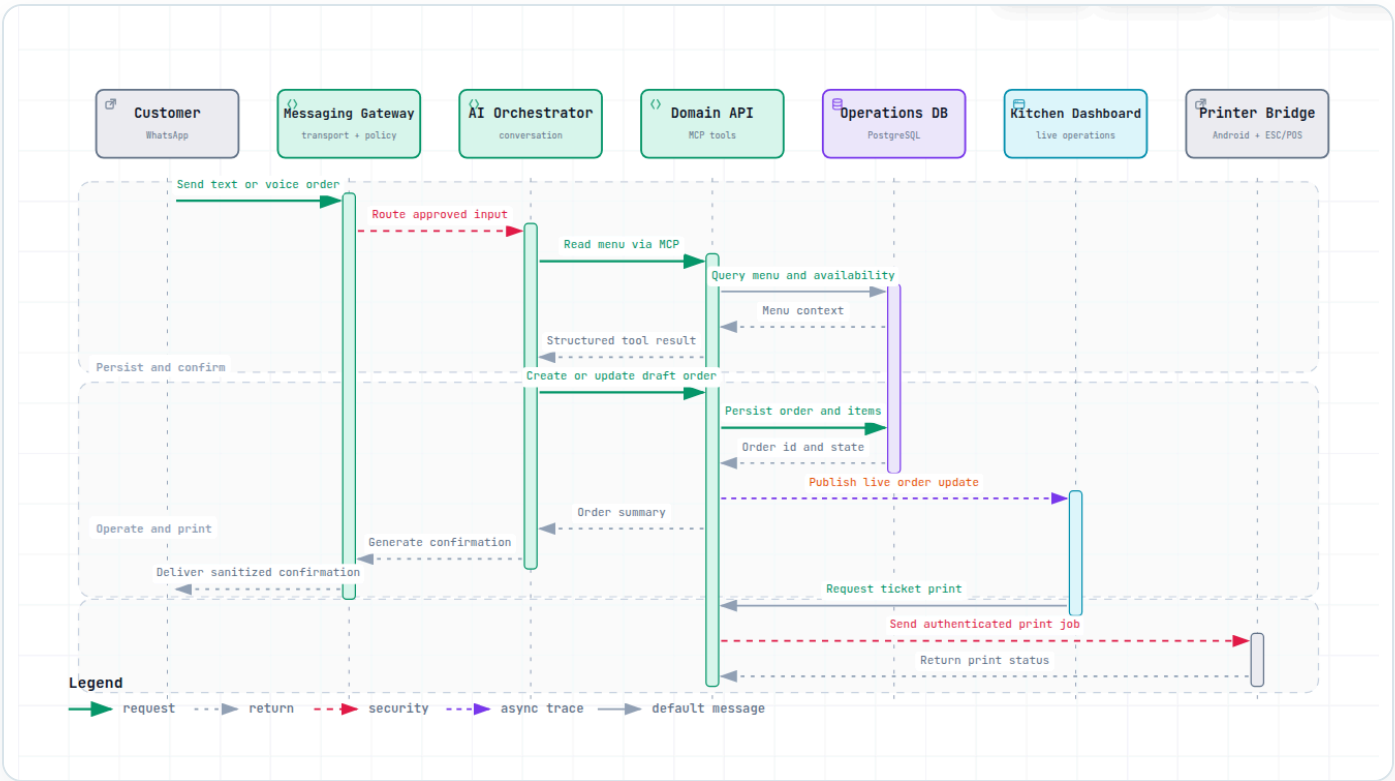
Printer bridge

Android and ESC/POS

EXECUTION VIEW

Order intake to fulfillment

A customer message becomes an operational order only after policy checks, current-menu lookup, structured tool execution, and database persistence. Dashboard and printer actions consume the persisted state rather than an unverified model response.



Representative happy path; operational retries and proprietary rules are intentionally omitted.

AI separated from business state

The model selects typed tools; domain validation and persistence remain deterministic services.

Confirmation after persistence

The customer receives a sanitized confirmation only after the order and items have a durable state.

Authenticated print dispatch

Print jobs cross a defined service boundary and return device status to the domain workflow.

Layered verification

Unit, integration, and end-to-end coverage protect the most failure-sensitive paths across APIs and devices.

Public-use note

This document demonstrates architecture thinking and delivery ownership without publishing the private repository or exposing implementation details owned by the employer.